

Live Twilio SMS Fallback Testing Report

OVERALL RESULT
6 / 6 PASS

Generated: 2026-06-12 · Automated test via `test_sms_constraint.py` · Gemini Flash API (Live)

FALLBACK ARCHITECTURE & HARD CONSTRAINTS

How the Offline Fallback Works

- User sends SMS**
User texts the Twilio-provisioned number from a basic keypad phone — no data balance, no WhatsApp required.
- Twilio routes to `/sms`**
The POST webhook hits the `/sms` FastAPI endpoint, not `/webhook`, activating SMS-only mode.
- LLM uses concise prompt**
An SMS-specific instruction is injected into the LLM context: *"ultra-concise, under 300 characters, no URLs."*
- Hard-enforcement sanitizer**
`sanitize_sms_reply()` strips URLs via regex and truncates to 300 chars — a code-level guarantee, not a prompt suggestion.

Enforced Constraints



≤ 300 Characters

Guarantees a single-part SMS. Multi-part messages (160 × N) create unexpected billing and may arrive out of order on legacy 2G networks.



Zero URLs / Hyperlinks

Keypad phones cannot tap links. All checklist download URLs are stripped from SMS replies automatically by the sanitizer.

CODE IMPLEMENTATION

`sanitize_sms_reply(text: str) → str`

Located in `backend/main.py` — called post-translation on every `/sms` response. Logs char savings to console.

AUTOMATED TEST RESULTS — FULL CONVERSATION TRANSCRIPT

6-Turn SMS Simulation (Interview → Profile Complete → Scheme Retrieval)

Simulated Sender: +91 98765 43210

#	USER MESSAGE	BOT REPLY (SMS)	CHARS	URLS	RESULT
1	Hello, I need help with government schemes	Welcome! To find the best schemes for you, may I know your age?	63/300	None	PASS
2	My age is 35	Thanks! May I know your gender?	31/300	None	PASS

#	USER MESSAGE	BOT REPLY (SMS)	CHARS	URLS	RESULT
3	I am a male farmer	What is your annual income or BPL/APL status?	45/300	None	PASS
4	My income is below poverty line	Which state in India do you live in?	36/300	None	PASS
5	I live in Uttar Pradesh	Thank you! I have successfully gathered your profile. Please reply with 'show schemes' to see your personalized recommendations!	128/300	None	PASS
6	show schemes	Schemes for you: 1. PM-KISAN (₹6k/yr) Why: You are a farmer. Docs: Aadhaar, Bank A/c, Land papers. 2. PMAY-G (House) Why: You are BPL. Docs: Aadhaar, Bank A/c, Job Card.	170/300	None	PASS

Full machine-readable log: [backend/test_log.txt](#) · Generated by live Gemini Flash API calls (not mocked)

TWILIO CONSOLE CONFIGURATION — LIVE DEPLOYMENT STEPS

1

Start Backend
Run the FastAPI server from the venv:

```
cd backend
.\venv\Scripts\uvicorn.exe main:app --reload
```

2

Expose via Tunnel
Use ngrok or localtunnel to create a public URL:

```
ngrok http 8000
# or:
lt --port 8000
```

3

Configure Twilio
Phone Numbers → Manage → Active Numbers → SMS number → Messaging → "A MESSAGE COMES IN"

```
https://<ngrok-url>
.ngrok-free.app/sms
```

4

Send Test SMS
Text the Twilio number from a physical keypad phone. Confirm: single bubble reply, no links, under 300 chars.